14 Bound States in External Fields

Problems

1. Consider a charged scalar particle of mass $m \neq 0$, described by a field $\phi(x)$ whose only interaction is with an external time-independent electromagnetic field $\mathcal{A}^\mu(x)$. Let $|N\rangle$ be a complete set of normalized one-boson or one-antiboson states with energies E_N, and define

$$u_N(\mathbf{x})e^{-iE_N t} = \langle\Omega|\phi(\mathbf{x},t)|N\rangle, \qquad v_N(\mathbf{x})e^{iE_N t} = \langle N|\phi(\mathbf{x},t)|\Omega\rangle,$$

   where $|\Omega\rangle$ is the vacuum. Show that the u_N and v_N together form a complete set, and give formulas for the coefficients of u_N and v_N in the expansion of a general function $f(\mathbf{x})$.

2. Suppose we include radiative corrections in the theory of Problem 1. Let $i\Pi^*(x,y)$ be the sum of all diagrams with one incoming and one outgoing charged scalar line (excluding final scalar propagators) to order α. Derive a formula for the shift in the energies E_N of the one-boson states due to these radiative corrections, in terms of $u_N(\mathbf{x})$ and $\Pi^*(x,y)$.

3. Use the results of Section 14.3 to calculate the radiative decay rate of the $2p$ states of hydrogen.

4. Suppose that the electron has an interaction with a light scalar field ϕ, of the form $g\phi\bar{\psi}_e\psi_e$. Suppose that the scalar mass m_ϕ is in the range $(Z\alpha)^2 m_e \ll m_\phi \ll Z\alpha m_e$. Calculate the change in the energy of the $1s$ state of hydrogenic atoms due to this interaction.

5. Carry out the calculation of Problem 4 for $m_\phi = 0$.

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